Summary

- For any natural number n, n factorial is the product of the first n natural numbers and is denoted by $n!$ or $\lfloor \underline{n} \rfloor$.
- For any natural number n, $n! = n(n-1)\,(n-2)\ldots 3\times 2\times 1$
- $0!=1$
- If there are two jobs, each of which can be performed independently in m and n ways respectively, then either of the two jobs can be performed in $(m+n)$ ways.
- The number of arrangements that can be made out of n things taking r at a time is called the number of permutation of n things taking r at a time.
- $np_r = \dfrac{n!}{(n-r)!}$
- $np_o = \dfrac{n!}{(n-0)!} = \dfrac{n!}{n!} = 1$
- $np_1 = \dfrac{n!}{(n-1)!} = \dfrac{n(n-1)!}{(n-1)!} = n$
- $np_n = \dfrac{n!}{(n-n)!} = \dfrac{n!}{0!} = n!$
- $np_r = n(n-1)(n-2)\ldots[n-(r-1)]$
- The number of permutation of n different things taken r at a time, when repetition is allowed is n^r.
- The number of permutations of n things taken all at a time, of which p things are of one kind and q things are of another kind, such that $p+q = n$ is $\dfrac{n!}{p!\,q!}$
- Circular permutation of n different things taken all at a time = $(n-1)!$
- The number of circular permutation of n identical objects taken all at a time is $\dfrac{(n-1)!}{2}$
- Combination is the selection of n things taken r at a time without repetition.
- Out of n things, r things can be selected in nc_r ways.
- $nC_r = \dfrac{n!}{r!(n-r)!},\ 0 \le r \le n$
- $nC_0 = nC_n = 1.$
- $nC_1 = n.$
- $nC_2 = \dfrac{n(n-1)}{2!}$
- $nC_x = nC_y$, then either $x = y$ or $x + y = n$.

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